

# RB-OPS-001：Kuboard连接Database异常Runbook

## Metadata

- 环境: 生产
- 服务名称: kuboard
- 告警名称: 连接异常
- 负责人: @sre-oncall
- 最后更新: 2026-07-30
- 下次演练: xxx
- 预计耗时: 5-20 分钟
- 严重级别: P1 (全量不可用)

## 触发条件

- ☒ 用户反馈：Web页面无法正常显示

## 前置检查清单

- ☒ 已连接进入ecs
- ☒ 进入docker部署的kuboard服务
- ☒ 已通知值班群：「<service-name> 系统繁忙故障排查中，@oncall」

## 快速诊断

### 步骤1：查看web报错

```
http://iot-amqp.tian-power.com/sso/auth?access_type=offline&client_id=kuboard-
sso&redirect_uri=%2Fcallback&response_type=code&scope=openid+profile+email+groups
&state=%2F&connector_id=default
{
  "message": "Failed to connect to the database.",
  "type": "Internal Server Error"
}
```

## 步骤2：查看容器kuboard报错

```
# 查看最近33条日志
docker logs --tail 33 kuboard
# 跟踪日志（会全部展示）
docker logs -f kuboard

time="2026-07-30T03:25:10Z" level=error msg="Storage health check failed: create
auth request: etcdserver: mvcc: database space exceeded"
{"level":"warn","ts":"2026-07-
30T11:25:25.311+0800","caller":"clientv3/retry_interceptor.go:61","msg":"retrying
of unary invoker failed","target":"endpoint://client-591df4e6-e4f3-4265-a930-
36d55795354c/127.0.0.1:2379","attempt":0,"error":"rpc error: code =
ResourceExhausted desc = etcdserver: mvcc: database space exceeded"}
time="2026-07-30T03:25:25Z" level=error msg="Storage health check failed: create
auth request: etcdserver: mvcc: database space exceeded"
```

## 步骤3：日志审批

```
# etcdctl --write-out=table endpoint status , # DB SIZE 为 2.1 GB（已超默认配额）。
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
|      ENDPOINT      |      ID      | VERSION | DB SIZE | IS LEADER | IS LEARNER |
| RAFT TERM | RAFT INDEX | RAFT APPLIED INDEX |      ERRORS      |
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
| 127.0.0.1:2379 | 59a9c584ea2c3f35 | 3.4.14 | 2.1 GB | true | false |
|         4 | 6081756 | 6081756 | memberID:6460912315094810421 |
|         |         |         |         |         |         |
|         |         |         |         | alarm:NOSPACE |
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
```

输出中 `alarm:NOSPACE` 明确表明空间已满

- 已在 etcd 所在服务器上，且有 `etcdctl` 可执行权限。
- etcd 的存储空间（默认 2GB）已写满，触发 **空间不足（No Space）** 告警，此时 etcd 会**拒绝所有写入操作**，只能读取

## 步骤4：处理故障

```
# 获取当前最新的 revision
ETCDCTL_API=3 etcdctl --endpoints=http://127.0.0.1:2379 endpoint status --write-out=json | egrep -o '"revision":[0-9]*' | egrep -o '[0-9].*'
6081501

# # 压缩历史版本（Compaction）etcd 会保留历史修订版本，压缩可释放空间
ETCDCTL_API=3 etcdctl --endpoints=http://127.0.0.1:2379 compact 6081501
compacted revision 6081501

# 碎片整理（Defragmentation）压缩后，磁盘空间不会立即归还，需要整理：
ETCDCTL_API=3 etcdctl --endpoints=http://127.0.0.1:2379 defrag

# 解除告警
ETCDCTL_API=3 etcdctl --endpoints=http://127.0.0.1:2379 alarm disarm
```

## 步骤5：验证

```
#确认 DB SIZE 下降，且 ERRORS 列为空
etcdctl --write-out=table endpoint status
```

| ENDPOINT       | ID               | VERSION | DB SIZE | IS LEADER | IS LEARNER |
|----------------|------------------|---------|---------|-----------|------------|
| 127.0.0.1:2379 | 59a9c584ea2c3f35 | 3.4.14  | 156 kB  | true      | false      |
| 4              | 6081804          | 6081804 |         |           |            |